

Park+ : Empowering drivers with an automotive super app

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

Park+ carried out complex migration in 2 months, improved homepage load time by 60%, gained up to 10% boost in new product launches, and deployed ML-driven expansion strategy with Google Cloud

ABOUT PARK PLUS



About Park Plus

Park+ is an automotive superapp on a mission to empower Indian drivers by helping them with all car-related needs. What started as a mission to connect commuters to safe, secure, and digitized parking spaces expanded to include services, such as recharging FASTag electronic wallets, finding car maintenance, locating the nearest and cheapest gasoline stations, buying and renewing insurance, comparing used cars, and more. As co-founders Amit Lakhotia and Hitesh Gupta connected with users on Park+, they identified the need to expand its horizons to

**Google
Cloud
results**

**More
hours**

- Enhances user experience by reducing platform latencies by 14%
- Supports mission to reach 60% of India's drivers with personalized solutions enabled by BigQuery data analytics
- Deepens customer loyalty with up to 5x boost to speed of new product launches
- Supports rapid growth trajectory by absorbing 2x growth in traffic every three months

engine
hours
year

operate under one simple rule of "simplifying the process of owning a car."

Industries: Travel & Hospitality

Location: India

(<https://cloud.google.com/kubernetes-engine/>),



About Niveus Solutions

Niveus Solutions is a Google Cloud Partner on a people-centric mission to enable Indian innovators from healthcare to finance and entertainment to optimize their digital transformation. In 2020, it was named Asia-Pacific Google Cloud Partner of the Year.

ABOUT NIVEUS

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Load Balancing

(<https://cloud.google.com/load-balancing/>)

named Asia-Pacific Google Cloud Partner of the Year.

For millions of people in India, a car is a tool of empowerment as they work toward enhancing their financial well-being. Yet myriad challenges arise in the car ownership experience, from choosing the right vehicle to locating parking spaces in congested cities, as well as getting expert maintenance and purchasing the best insurance.

As such, **Park+** (<https://parkplus.io/>) is a super app for all automotive needs on a mission to "simplify the process of owning a car" by offering a data-driven ecosystem of car-related services. Beginning as an app to connect drivers to secure parking, it tackled common pain points faced by drivers to become India's digital one-stop-shop for their convenience. This includes a trademark automatic top-up system for contactless payments, such as electronic toll collection, shopping mall parking, and gasoline filling.

Park+ started life as a microservices-based architecture built on a variety of open-source software that enabled the platform to grow rapidly in its early stages. As the solution's popularity soared across India, the startup found itself struggling to keep up with demand. Its small DevOps team faced a growing burden in manual configuration of instances, while tending to chores, such as load testing and security patching. With new users reaching into the millions, all requiring a variety of real-time services, the platform began to encounter issues with latency and scalability that affected customer experience.

"In our mission to empower Indian drivers, we're finding traffic doubling every three to six months, and manual provisioning became overwhelming. By turning to a combination of Google Kubernetes Engine, Cloud Load Balancing and BigQuery we gained the absolute freedom to scale our platform and delight customers with tailored solutions."

—Hitesh Gupta, Co-founder and CTO, Park+

Working with partner Niveus Solutions, Park+ decided to embark upon a complex migration to Google Cloud (<https://cloud.google.com/>) thanks to what it considered a best-in-class, fully-managed microservices solution in Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) that would enable autoscaling to petabytes of data.

Park+ was also drawn to the power of Cloud Load Balancing

(<https://cloud.google.com/load-balancing?hl=en>) to direct platform traffic during usage spikes, and BigQuery (<https://cloud.google.com/bigquery>) for autoscaling data analytics of user behavior to enable a continuous stream of personalized offerings.

Thanks to proactive support from Niveus Solutions (<https://niveussolutions.com/>) and the Google Cloud team, Park+ was able to carry out the multi-phased migration with zero platform downtime in under two months, instead of the expected three. Google Cloud adoption enabled seamless growth to 10 million platform users within a year, while freeing the engineering team from mundane tasks to focus on innovation.

"In our mission to empower Indian drivers, we're finding traffic doubling every three to six months, and manual provisioning became overwhelming," says Hitesh Gupta, Co-founder and CTO, Park+. "By turning to a combination of Google Kubernetes

Engine, Cloud Load Balancing and BigQuery we gained the absolute freedom to scale our platform and delight customers with tailored solutions."

Driving the future of India's automotive journey via a partnership

Gupta recalls the day Park+ went live on its new Google Cloud infrastructure after an intensive eight weeks working with Niveus Solutions on testing and transferring services to the new architecture. Going beyond being a cloud vendor, the Google Cloud team was on site with Park+ throughout the night to support the migration, helping with load and application testing to ensure the launch went glitch-free.

"Alongside Niveus Solutions, the supportive experience the Google Cloud team gave us was truly exceptional and something I've never seen on any other platform," says Gupta. "They stayed up with us all night on the D-Day of migration, showing a strong partnership model that helped the complex migration go with seamless perfection."

"Alongside
Niveus
Solutions, the
supportive

experience the
Google Cloud
team gave us
was truly
exceptional and
something I've
never seen on
any other
platform. They
stayed up with
us all night on
the D-Day of
migration,
showing a
strong
partnership
model that
helped the
complex
migration go
with seamless
perfection."

*—Hitesh Gupta, Co-
founder and CTO, Park+*

Since that day, Park+ has achieved impressive
results with Google Cloud, as it powers its mission

to rapidly move beyond a community of 10 million drivers, or 35% of the Indian driving population, en route to a goal of reaching 60% of India's drivers. With the autoscaling capabilities of GKE and platform agility and stability enabled by Cloud Load Balancing, Park+ has achieved a 14% improvement in platform latencies from 100 milliseconds to 88 milliseconds. Homepage load time has gone down from 250 milliseconds to 100 milliseconds. This has translated into significant improvement in user experience and website stickiness, which facilitated its business growth.

Gupta says GKE has streamlined the process of deploying new services that are critical to the Park+ mission as a constantly evolving super app. By managing the entire cycle of application deployment, GKE has enabled the platform to reduce the time needed to launch new features from up to five minutes, to between one and three minutes for any upgrade, no matter how large the load.

"By virtually
every measure,
our cloud
infrastructure
built on Google
Kubernetes
Engine and

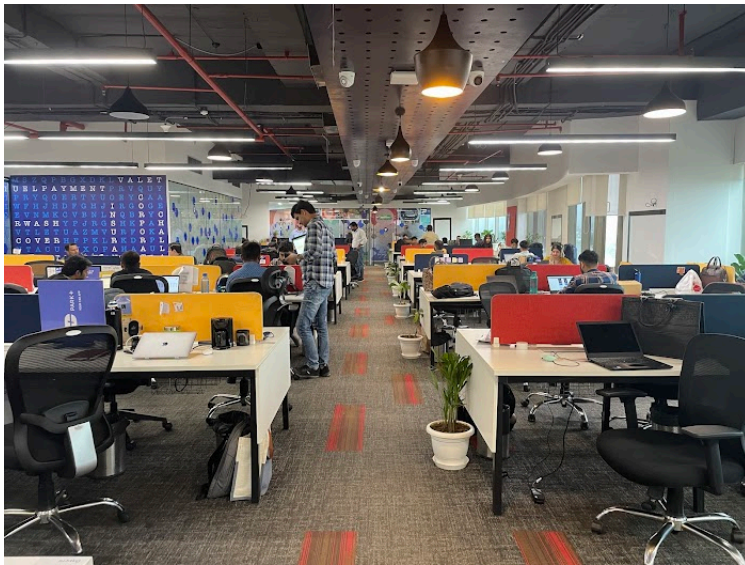
Cloud Load
Balancing to
direct platform
traffic has
enabled the
explosive growth
of our platform.

Most
importantly,
these powerful
managed
services are
freeing our team
to help recreate
the journey of
Indian mobility."

*—Hitesh Gupta, Co-
founder and CTO, Park+*

Meanwhile, by freeing the technology team from repetitive tasks, Park+ has reduced the number of engineers dedicated to computing infrastructure from five to two. Overall, Gupta says Park+ is saving more than 900 hours per year in technology team hours, since the days it was managing an open source infrastructure. These are time and resources that can now be devoted to creative problem solving.

"By virtually every measure, our cloud infrastructure built on Google Kubernetes Engine and Cloud Load Balancing to direct platform traffic has enabled the explosive growth of our platform," says Gupta. "Most importantly, these powerful managed services are freeing our team to help recreate the journey of Indian mobility."



Park Plus' office

Driving an endless future of car owner empowerment

The impressive growth that Park+ is achieving, thanks to such platform agility and limitless autoscaling, is bringing in troves of consumer data, from driving habits to parking locations and automobile maintenance work. All of this driver data, secured by Google Cloud built-in data protection protocols, can be fed through BigQuery to unlock pinpointed insight for the development team to improve services.

On the road to growing to 60% of the Indian driving public, Gupta says Park+ is looking to a future of even more targeted data insight from its community to drive machine learning-enabled personalized solutions. The platform features a community forum where users discuss car problems, share solutions and driving experiences, and post selfies next to their automobile.



Park Plus' employees engrossed in work

These interactions will enable Park+ to evolve into an AI-enabled recommendation engine for a holistic driving experience, including places where drivers might like to go shopping, have a meal, or stay for the night. For this next stage of the platform evolution, Gupta envisions possible uses for Google Cloud AI and ML solutions.

Park+ is now setting its sights on global expansion based on the insights it has gained from solving driver challenges and pain points in India. Gupta

says that the world-class network of data centers by Google Cloud, and its services and support network in every market, promises to make the expansion endlessly dynamic.

"We're excited about bringing the solution we've built for more than 10 million Indian drivers to a global community of car lovers," says Gupta. "I'm certain that Google Cloud being available everywhere with a world-class support team and powerful solutions will allow us to launch fast and seamlessly in exciting new markets."